

Student Performance Prediction using Machine Learning

Submitted By

Name: Priya Shah

Domain: Data Science & Machine Learning

Abstract

This project develops a beginner-friendly machine learning system to predict students' final examination scores using study hours, attendance, assignment marks, and previous examination scores. The project follows the complete data science lifecycle from data collection and preprocessing to visualization, model training, evaluation, and prediction. It demonstrates how educational data can be analyzed to support informed academic decisions.

Introduction

Data Science combines statistics, programming, and domain knowledge to extract useful insights from data. Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from historical data. Educational institutions increasingly use data-driven techniques to improve student outcomes. Predicting academic performance helps teachers identify students who require additional attention and enables timely intervention.

Problem Statement

Student performance depends on multiple factors such as attendance, study habits, assignments, and previous academic records. Manually analyzing these factors for every student is difficult. Therefore, this project aims to build a simple predictive model that estimates a student's final score from available academic information.

Objectives

1. Learn the complete Data Science workflow.
2. Perform data preprocessing.
3. Explore the dataset using statistical methods and graphs.
4. Train a Linear Regression model.
5. Evaluate the model using MAE, MSE and R^2 Score.
6. Develop a Streamlit interface for easy predictions.

Tools & Technologies

Python was used as the programming language. Pandas handled data manipulation, NumPy supported numerical operations, Matplotlib and Seaborn created visualizations, Scikit-learn built the machine learning model, Joblib saved the trained model, Streamlit created the user interface, and GitHub was used for version control.

Dataset Description

The dataset contains Study Hours, Attendance, Assignment Score, Previous Score, and Final Score. These features represent a student's academic activities and are used to train the prediction model. The target variable is the Final Score.

Methodology

The workflow included importing the dataset, checking for missing values and duplicates, performing exploratory data analysis, selecting features, splitting the data into training and testing sets, training the Linear Regression model, evaluating predictions, saving the trained model, and creating a web application for users.

Exploratory Data Analysis

EDA included histogram, scatter plot, box plot, correlation heatmap, and pair plot. The analysis showed that study hours and previous examination scores had a strong positive relationship with final scores. Attendance also contributed positively to student performance.

Model Development

Linear Regression was selected because it is easy to understand and suitable for predicting continuous numerical values. The dataset was divided into training and testing data using an 80:20 split. After training, the model predicted unseen student scores.

Results & Discussion

The model produced satisfactory predictions for a beginner-level dataset. Evaluation metrics such as Mean Absolute Error, Mean Squared Error, and R^2 Score were used to measure model performance. The results indicate that academic factors included in the dataset significantly influence final examination scores.

Advantages

Simple implementation, easy interpretation, low computational cost, suitable for beginners, demonstrates the complete ML workflow, and can assist educational institutions.

Limitations

The project uses a relatively small dataset and only one algorithm. Real-world accuracy can be improved with larger datasets, additional features, and advanced machine learning models.

Future Scope

Future enhancements include larger datasets, feature engineering, Random Forest and XGBoost models, cloud deployment, real-time dashboards, and integration with school management systems.

Conclusion

This project successfully demonstrates a complete beginner-friendly machine learning pipeline. It strengthened understanding of data preprocessing, visualization, statistical analysis, model building, evaluation, and deployment. The project serves as a strong foundation for learning advanced Data Science and Machine Learning concepts.

GitHub Repository

https://github.com/Priyashah15/Student_Performance_Prediction

References

[Python Documentation](#), [Pandas Documentation](#), [NumPy Documentation](#), [Scikit-learn Documentation](#), [Matplotlib Documentation](#), [Seaborn Documentation](#), [Streamlit Documentation](#), [GitHub Repository](#).